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Academic Year: 2023-2024

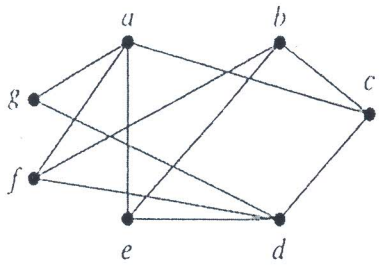
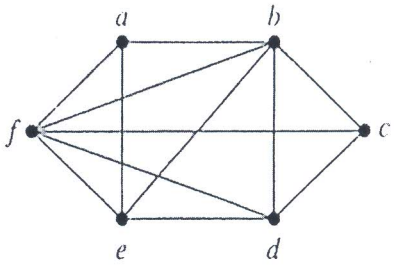
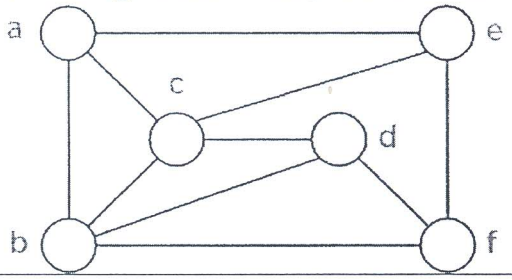
Programs: B. Tech / MBA Tech Streams: DS/AIDS/AIML/AI/CSE(CS)/CS/CE/ETE/EXTC/IT	Year: 2nd /3rd Semester: 3rd /5th
Subject: Discrete Mathematics	Time: 3 hrs. (10 am to 1 pm)
Date: 23/11/2023	No. of Pages: 3
Marks: _100_____	

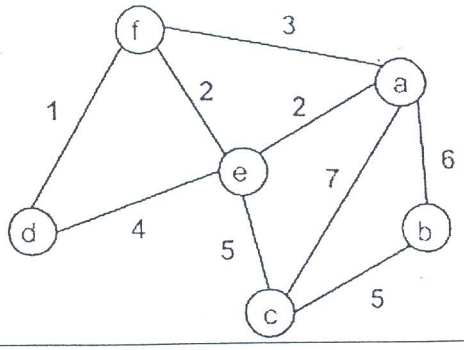
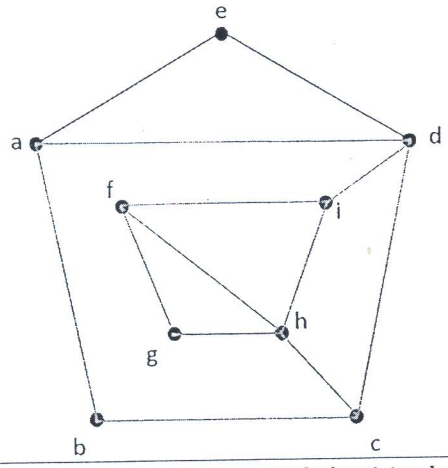
Final Examination/Re-Examination (~~2021-22~~, 2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all __5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

CO-1 ; SO- 2; BL-3	Q.1 a)	Draw a single Venn diagram to represent the case $A \subseteq C, B \subseteq C$. Hence or otherwise draw another Venn diagram to represent the case $C \subseteq A, C \subseteq B$.	4
CO-2 ; SO- 1; BL-2	b)	Show that the statement $((p \vee q) \wedge (p \rightarrow q)) \vee \overline{q}$ is a tautology.	4
CO-1 ; SO- 1; BL-1	c)	20 cards numbered 1 to 20 are placed face down on a table. Cards are selected at random one at a time and turned over until 11 cards have been chosen. If two of the cards add up to 21, the player loses. Is it possible to win this game?	4
CO-4 ; SO- 1; BL-3	d)	Identify: i) for which values of n are the graphs K_n bipartite? ii) for which values of n are the graphs K_n regular?	4
CO-4 ; SO-2 ; BL-5	e)	Determine whether the set of odd integer together with the operation $a * b = ab$ is a monoid. Justify.	4
CO- 3; SO- 2; BL-5	Q.2 a)	Determine the solution for recurrence relation $a_n = 2a_{n-1} + a_{n-2}$ for $n \geq 2$ where $a_0 = 0$ and $a_1 = 1$.	6
CO- 1; SO- 2; BL-3	b)	Are the graphs G and H displayed below bipartite? Justify your answer. If bipartite redraw the graph showing the partitions.	6

		 G  H	
CO-3 ; SO-2 ; BL-4	c)	Let $\mathbb{Z}$ denotes the set of integers. Define a relation R on $\mathbb{Z}$ by setting xRy if and only if $x = y$ or $x = -y$. (i) Examine whether R is equivalence relation or not. (ii) Find all equivalence classes of $\mathbb{Z}$ induced by R.	8
CO-2 ; SO- 1; BL-2	Q.3 a)	By mathematical induction, show that $$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$	6
CO- 2; SO- 2; BL-6	b)	Use the Welsh-Powell algorithm to find the chromatic number of the following graph. 	6
CO- 1; SO- 1; BL-4	c)	Let $A = \{1,2,3,4\}$. Examine whether the relation $R = \{(1,1),(1,2),(2,1),(2,2),(3,3),(3,4),(4,3),(4,4)\}$ is symmetric, asymmetric, antisymmetric and transitive. Give argument in support of your answer.	8
CO- 3; SO- 2; BL-4	Q.4 a)	Show that the set of real numbers ($\mathbb{R}$) form an abelian group with respect to binary operation '$*$' defined as $a * b = a + b + 2,\ \forall a, b \in \mathbb{R}$.	6
CO-2 ; SO-2 ; BL-2	b)	Discuss the satisfiability and validity of the following argument: *If you have access to the network, then you can change your grade. You have access to the network. Therefore, you can change your grade.*	6
CO- 3; SO- 1; BL-5	c)	Determine the solution of the following recurrence relation: $a_n - 2\,a_{n-1} = 3 \cdot 2^n$, where $n \geq 1$ *and* $a_0 = 1$.	8
CO- 2; SO- 1; BL-4	Q.5 a)	Prove that the additive group of $(\mathbb{Z}_6, +_6)$ is isomorphic to multiplicative group of $(\mathbb{Z}_7^*, \times_7)$. Where $\mathbb{Z}_6 = \{0,1,2,3,4,5\}$ and $\mathbb{Z}_7^* = \{1,2,3,4,5,6\}$.	6
CO-3 ; SO- 1; BL-4	b)	Prove the following statement with clearly indicate the method of proof used. *If $5n + 3$ is odd, then $7n + 4$ is even*	6

CO- 3; SO- 1; BL-3	c)	Apply Prim's Algorithm to find the minimal spanning tree of the weighted graph: 	8
CO-1 ; SO- 2; BL-4	Q.6 a)	Determine whether the function $f: \mathbb{R} - \{7\} \longrightarrow \mathbb{R} - \{1\}$ defined as $f(x) = \frac{x+3}{x+7}$ is bijective, if it is then find f^{-1}.	6
CO- 4; SO- 1; BL-3	b)	Find the vertex connectivity and edge connectivity of the following graph. Justify your answer. 	6
CO- 4; SO- 1; BL-6	c)	If the addition and multiplication modulo 6 is defined on the set $R = \{0, 1, 2, 3, 4, 5\}$ then test that algebraic structure $(R, +_6, \times_6)$ is a ring with unity. Is it a field?	8
CO- 2; SO- 2; BL-3	Q.7 a)	Apply laws of logic to establish the equivalence: $p \vee q \vee (\sim p \wedge \sim q \wedge r) \equiv p \vee q \vee r$	6
CO-3 ; SO- 7; BL-3	b)	Let $G = \{0, 1, 2, 3, 4, 5\}$ set of congruence classes under addition modulo 6 $(+_6)$, (i) Construct composition table with respect to '$+_6$'. (ii) Identify Identity element of $(G, +_6)$. (iii) Identify the inverse of all the elements under $+_6$.	6
CO- 4; SO- 1; BL-2	c)	Explain Euler Circuit and Hamilton Circuit. Show whether the graph $K_{3,3}$ has Euler Circuit and Hamilton Circuit?	8